Data management
based on Metadata

Maggot

Metadata aggregation on data storage

Daniel Jacob - Maggot – INRAE

Data Management Plan

Resources

How does the management of data is it funded, especially in the long term?

Responsibilities in the project

What does the project consist of?
Who are the partners?
What policy on data management?
Who is responsible for the management of data?

Data collection

What data will be produced/used during the course of the project (type, format, volume and increase...) ?
How will they be produced?

Intellectual Property

Who will be the owner of the data produced?
How will they be used?

Data backup

How, where, by whom, will be stored, backed up and secured the data?

Data Access and Data sharing

Who will be able to access the data? The data will they be shared? published? With whom? How? How long does it take? Under which license?

Data Documentation

How will the data be identified, described? What metadata standards will be used?
How will the metadata be generated?

Data Archiving

What is the plan for long-term archiving and preservation?

Planning must be followed by implementation and therefore concrete actions

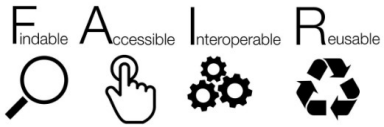
Data management

Being able to easily describe your data (descriptive metadata)

- with its professional vocabulary
- without tedious entries
- without having to re-enter the same information each time
- by associating external resources (links)

Being able to easily manage your data

- by limiting data loss (after the departure of temporary staff)
- by sharing only metadata
- be able to easily find data (from metadata)
- be able to provide access to data if necessary
- be able to distribute them without having to re-enter everything



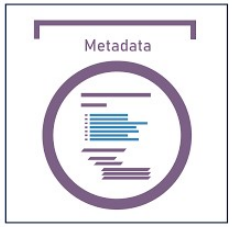
Ensuring metadata follows FAIR principles

- Respect a standard (metadata schema)
- Use controlled vocabulary consistent with your domain (thesaurus, ontologies)
- Be at least “Findable, Accessible & Interoperable”

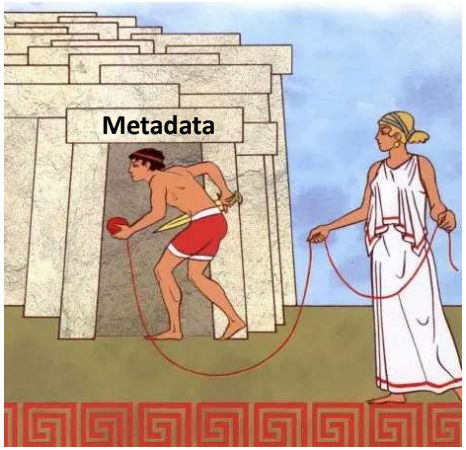
Data management

The Ariadne's thread …

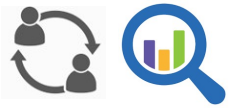
Organize
Your workspace
(storage, backup, naming, …)



Describe your data
(metadata)



Share & Search data/metadata
(metadata)



Publish your data
(with metadata)

Data management

The Ariadne's thread …

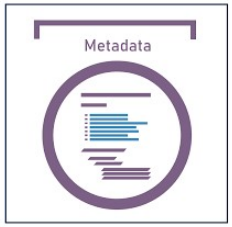
Organize
Your workspace
(storage, backup, naming, …)

Share & Search data/metadata
(metadata)

Publish your data
(with metadata)

Standards & Terminologies
4 FAIR pillars

Guidelines | Terminologies | Identifiers | Formats



Describe your data
(metadata)

Metadata schema

The Dataverse Project | zenodo

(what to describe)

Controlled Vocabulary

OntoPortal | SKOSMOS

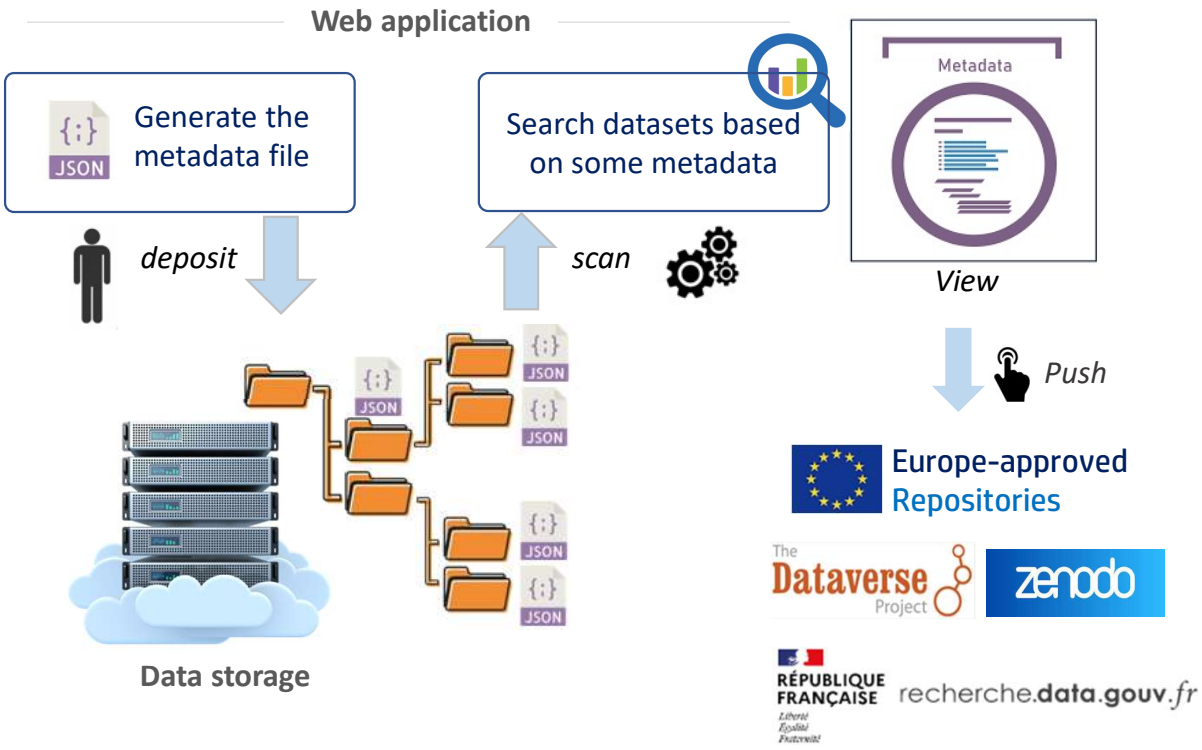
Dictionaries

AgroPortal, BioPortal,
VOINRAE, LOTERRE, ONTOSTACK,
…

(How to describe)

Maggot
Metadata aggregation on data storage

An ecosystem for metadata management



Describe metadata
with controlled vocabulary

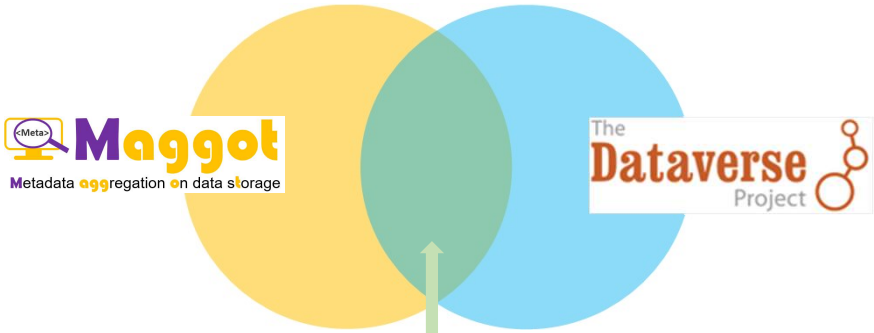
Share & Search metadata / data

Publish metadata
... with data

Describe metadata

Metadata Schema

Adoption of the Common Metadata of the schema implemented within the Harvard Dataverse software



Metadata schema (standard)

Data Document Initiative

DEFINITION
MANAGEMENT
STATUS
DESCRIPTORS
OTHER
...

Common Metadata
title, description, alternativeURL, contacts, authors, collectors, curators, members, depositor, leader, subject, keywords, topics, kindOfData, dataOrigin, lifeCycleStep, publication, otherReferences, grantNumbers, project, …

Specific Profile Metadata
...

Life Sciences Metadata
Geospatial Metadata
Journal Metadata
Semantic resource
...

Maggot
Metadata aggregation on data storage

Describe metadata

Guidelines
Metadata Schema

DEFINITION *
STATUS
MANAGEMENT *
DESCRIPTORS *
OTHER
RESOURCES

Short name *

Full title *

Subject *
Agricultural Sciences | Arts and Humanities | Astronomy and Astrophysics | Business and Management | Chemistry
Computer and Information Science | Earth and Environmental Sciences | Engineering | Law | Mathematical Sciences
Medicine Health and Life Sciences | Other | Physics | Social Sciences

Description of the dataset *

* mandatory fields

DEFINITION *
STATUS
MANAGEMENT *
DESCRIPTORS *
OTHER
RESOURCES

Kind of Data *
Audiovisual | Collection | Dataset | Event | Image | Interactive Resource | Model | Other | Physical Object
Service | Software | Sound | Text | Workflow

Keywords
Search a value: enter the first letters

Topic Classification
Search a value: enter the first letters

Data origin
Other | aggregate data | analysis data | audiovisual corpus | computer code | experimental data | observational data
simulation data | survey data | text corpus

Common Metadata of « Dataverse »

Common Metadata
title, description, alternativeURL, contacts, authors, collectors, curators, members, depositor, leader, subject, keywords, topics, kindOfData, dataOrigin, lifeCycleStep, publication, otherReferences, grantNumbers, project, …

Mandatory fields
Recommended fields
Desirable fields

Specific Profile Metadata

Experimental Factor
Search a value: enter the first letters

Measurement type
Search a value: enter the first letters

Technology type
Search a value: enter the first letters

Maggot
Metadata aggregation on data storage

Controlled vocabulary

Terminologies
Controlled Vocabulary

List of well-chosen and limited CVs (according to a reference e.g. Data Document Initiative)

Kind of Data *
☐ Audiovisual ☐ Collection ☐ Dataset ☐ Event ☐ Image ☐ Interactive Resource ☐ Model ☐ Other ☐ Physical Object
☐ Service ☐ Software ☐ Sound ☐ Text ☐ Workflow

Keywords

AgroPortal
BioPortal

List of ontologies to choose according to your domain

Search a value: experimental

- **Experimental** Model of Disease (NCBITAXON)
- **experimentally** modified cell in vitro (OBI)
- **experimental**_feature (OBI)
- **experimental** infection of cell culture (OBI)
- **experimental** disease induction (OBI)
- **Experimental** measurement (EDAM)

Topic Classifica

Search a value:

Use of dictionaries to target the CV by mixing thesaurus and ontologies

NAME (*)	ONTOLOGY	URL	Add new
NMR spectroscopy assay	OBI	http://purl.obolibrary.org/obo/OBI_0000623	Edit Del
agricultural science	EDAM	http://edamontology.org/topic_3810	Edit Del
amino acid	IOBC	http://purl.jp/bio/4/id/200906089657456524	Edit Del
analyte assay	MS	http://purl.obolibrary.org/obo/OBI_0000443	Edit Del
biochemical analysis	IOBC	http://purl.jp/bio/4/id/200906072808564316	Edit Del
biochemical characterization	IOBC	http://purl.jp/bio/4/id/201306093820876862	Edit Del
biochemical composition	IOBC	http://purl.jp/bio/4/id/201106016579695836	Edit Del
biochemistry	EDAM	http://edamontology.org/topic_3292	Edit Del

Thesaurus SKOSMOS

(VOINRAE, LOTERRE, ONTOSTACK, ...)

V.O.INRAE VOCABULAIRES OUVERTS

Building a vocabulary

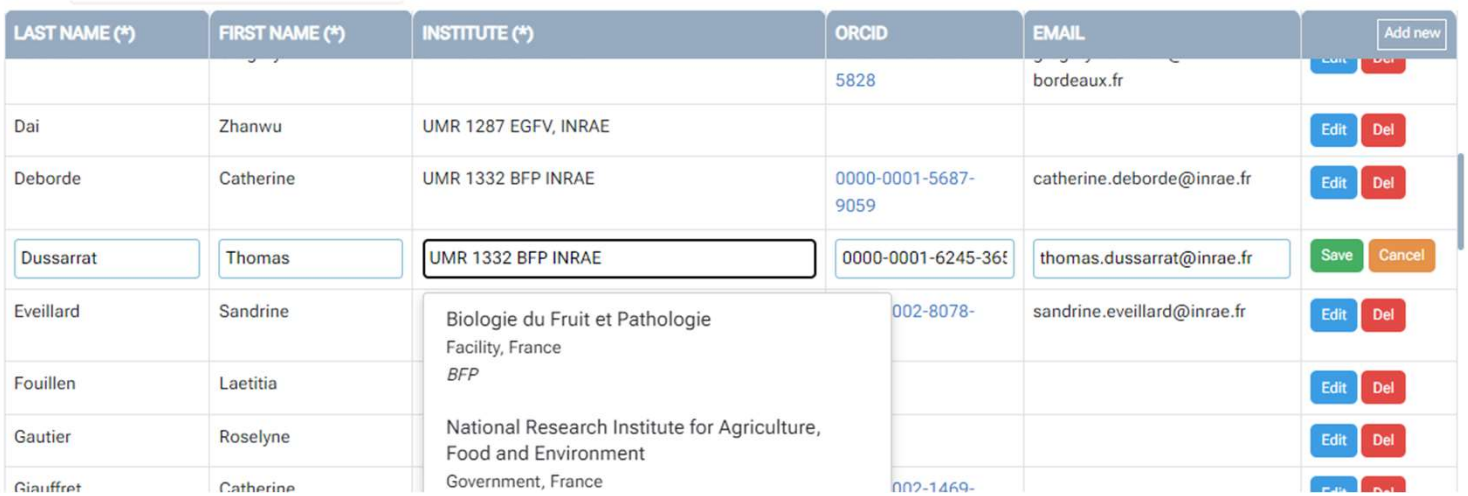
https://vocabulaires-ouverts.inrae.fr/construire/

Maggot
Metadata aggregation on data storage

Dictionaries

people, grant, producer, vocabulary, …

The use of **dictionaries within Maggot** has no other purpose to facilitate the entry of metadata, entry which can be long and repetitive in generalist data warehouses (such as repository based on Dataverse).



Dictionaries allow you to record multiple information necessary to define an entity, such as the names of people, or even the funders.

Its information, once entered and saved in a file called a dictionary, can be subsequently associated with the corresponding entity.

Maggot
Metadata aggregation on data storage

Dictionaries

Example : people

Contacts

Canlet Cécile, Deborde Catherine

Add a value: enter the first three letters

Authors

Canlet Cécile

Add a value: de

- Giraudeau Patrick
- Deborde Catherine
- Cahoreau Edern

Project Lead

Add a value: enter the first three letters

Data Collector

Add a value: enter the first three letters

Data Curator

Add a value: enter the first three letters

Data Member

Add a value: enter the first three letters

Push

Contact Use email button above to contact.

Canlet, Cécile (INRAE)
Deborde, Catherine (INRAE)

Author
Canlet, Cécile (INRAE) - ORCID: 0000-0002-6389-0712
Deborde, Catherine (INRAE) - ORCID: 0000-0001-5687-9059

Contributor
Project Leader : Giraudeau, Patrick (Univ. Nantes) - ORCID: 0000-0001-9346-9147
Data Collector : Deborde, Catherine (INRAE) - ORCID: 0000-0001-5687-9059
Data Collector : Canlet, Cécile (INRAE) - ORCID: 0000-0002-6389-0712
Data Collector : Gautier, Roselyne (INRAE)
Data Collector : Jousse, Cyril (Univ. Clermont Auvergne) - ORCID: 0000-0002-5899-8243
Data Collector : Lacaze, Mélia (INRAE)
Data Collector : Martineau, Estelle (Univ. Nantes) - ORCID: 0000-0001-5093-2138
Data Collector : Peyriga, Lindsay (INRAE) - ORCID: 0000-0002-6138-7961
Data Collector : Richard, Tristan (Univ. Bordeaux) - ORCID: 0000-0002-5308-8697
Data Collector : Silvestre, Virginie (Univ. Nantes)
Data Collector : Traïkia, Mounir (Univ. Clermont Auvergne) - ORCID: 0000-0002-4595-0400
Data Curator : Deborde, Catherine (INRAE) - ORCID: 0000-0001-5687-9059
Data Curator : Canlet Cécile (INRAE) - ORCID: 0000-0002-6389-0712
Data Curator : Moing, Annick (INRAE) - ORCID: 0000-0003-1144-3600
Data Curator : Jacob, Daniel (INRAE) - ORCID: 0000-0002-6687-7169
Project Member : Cahoreau, Edern (INRAE) - ORCID: 0000-0001-8637-0448
Project Member : Da Costa, Grégory (Univ. Bordeaux) - ORCID: 0000-0002-0336-5828
Project Member : le Mao, Inès (Univ. Bordeaux)

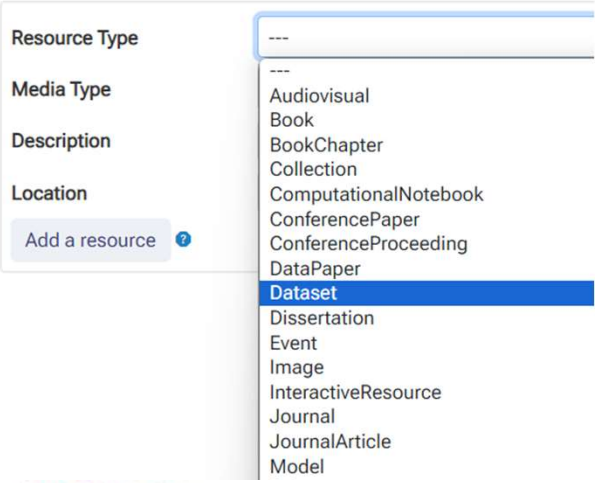
Thus, entering (by autocompletion) just the name of a person will allow the ORCID number, email address and institutional assignment to be associated when distributing metadata in Dataverse for example.

Maggot
Metadata aggregation on data storage

External Resources

"Data Fragmentation"

Identifiers ID — Persistent and unique identifiers



RESOURCES

Type	Media	Description	Location
JournalArticle		Journal of Experimental Botany, Oxford University Press, 2020	http://doi.org/10.1093/jxb/eraa302
Collection		ODAM Experimental data tables	https://pmb-bordeaux.fr/dataexplorer/?dc=Frimouss
Report	application/pdf	Fruit Growth Modelling	https://pmb-bordeaux.fr/getdata/pdf/Frimouss/FruitGrowthModelling.pdf
Software		Growth modeling applied to several fruit species	https://github.com/djacob65/growthmodel

We can also define external resources (URL links) relating to documents, publications or other related data.
Maggot thus becomes a hub for your datasets connecting different resources, local and external.

Maggot

Metadata aggregation on data storage

Formats

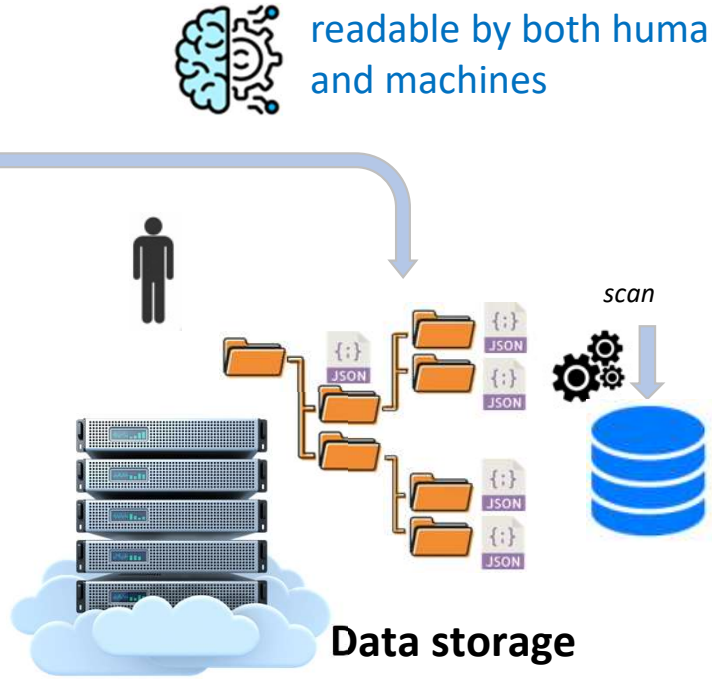
Format for interoperability purposes.

As output we produce a file in the format **JSON**

readable by both humans and machines

JSON

Metadata



Data storage

Local (meta)data repository

Storage space becomes the data repository

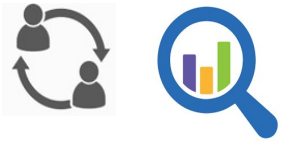
Infrastructure

Local, Remote or Mixte

Maggot

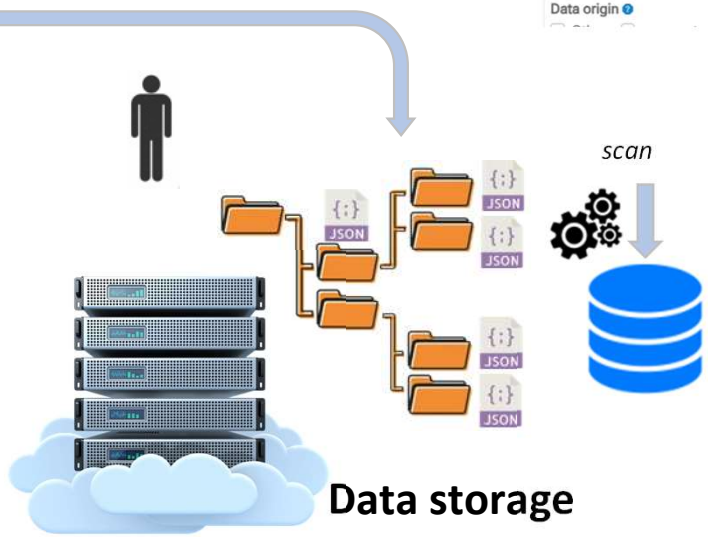
Metadata aggregation on data storage

Share & Search
(meta)data



JSON

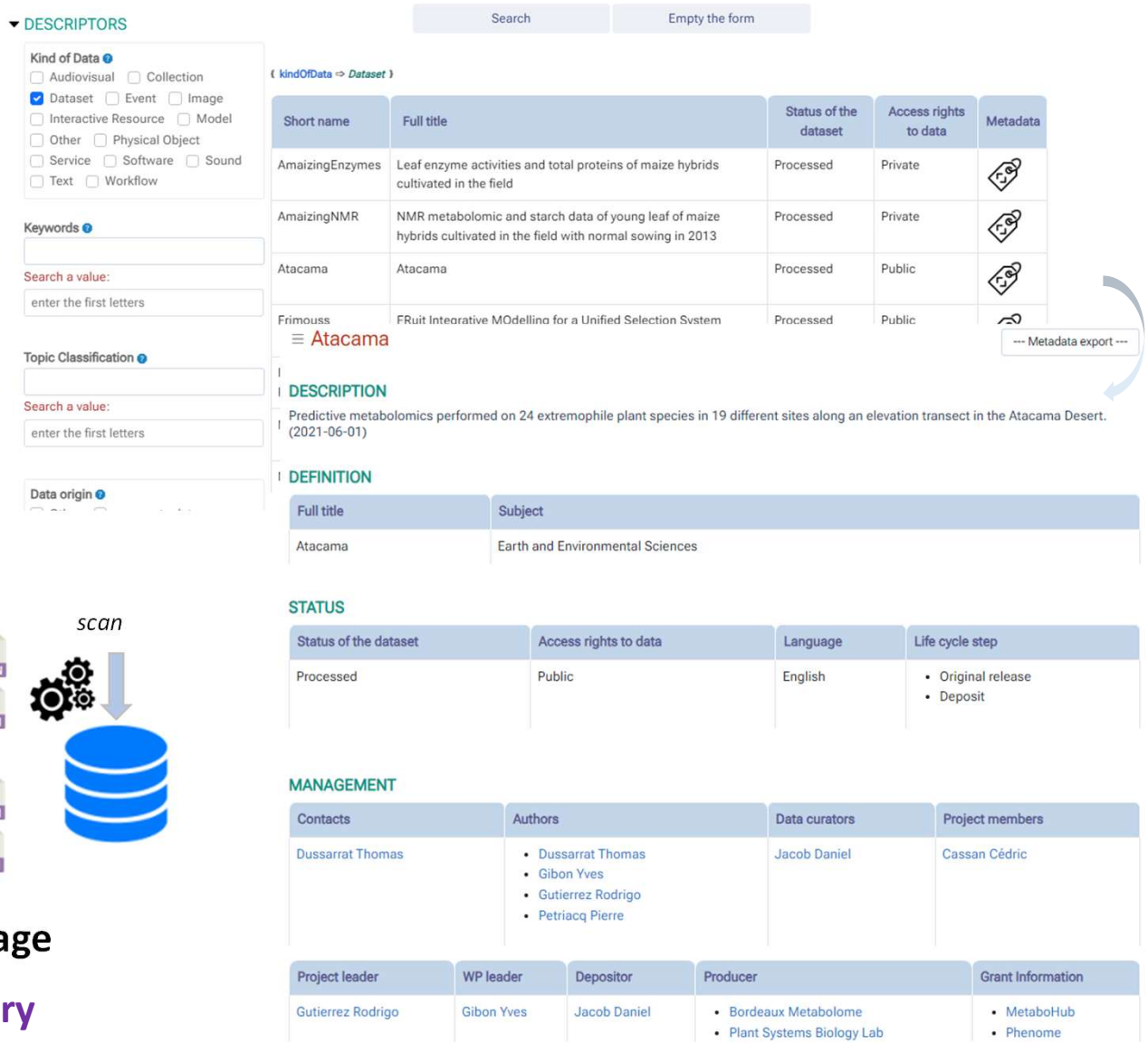
Metadata



Data storage

Local (meta)data repository

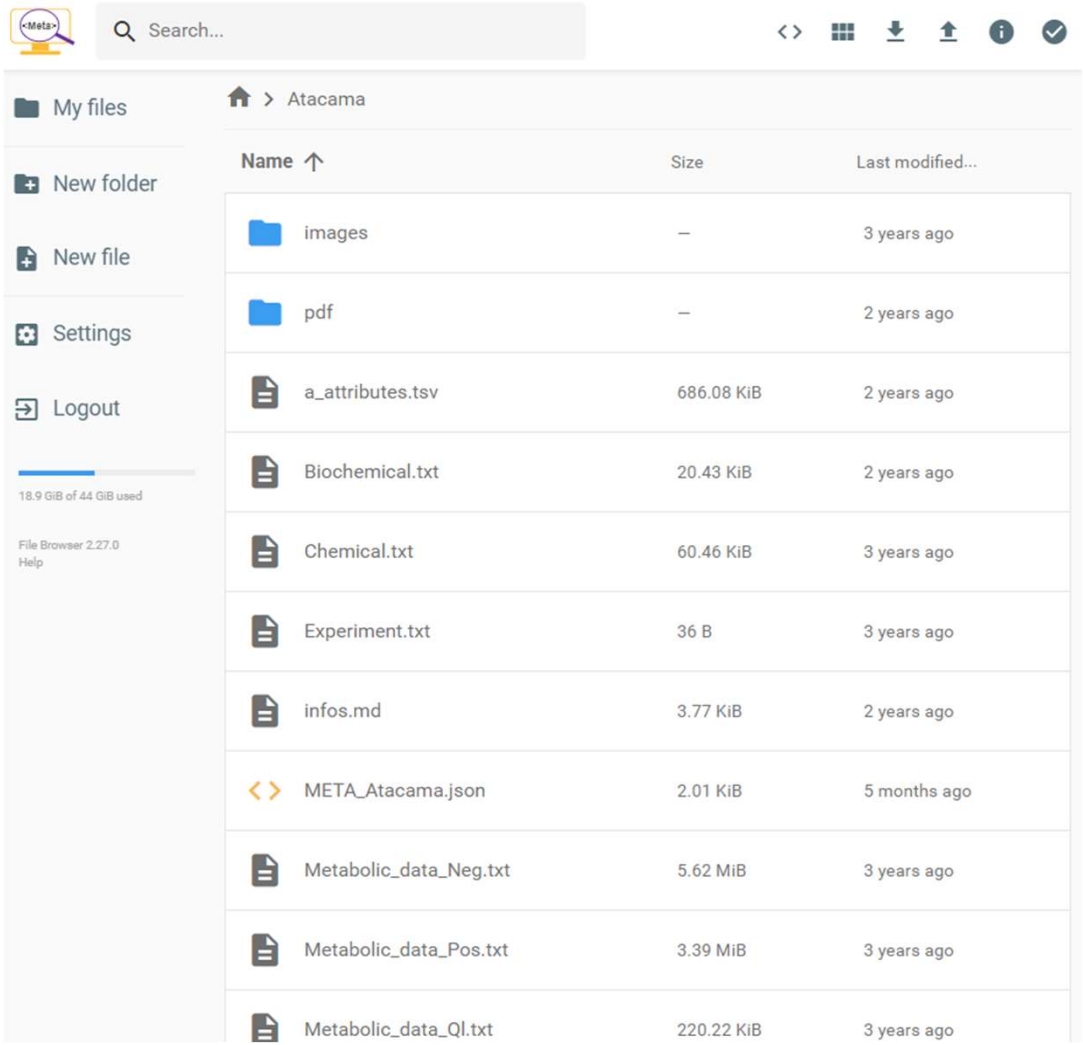
Storage space becomes the data repository

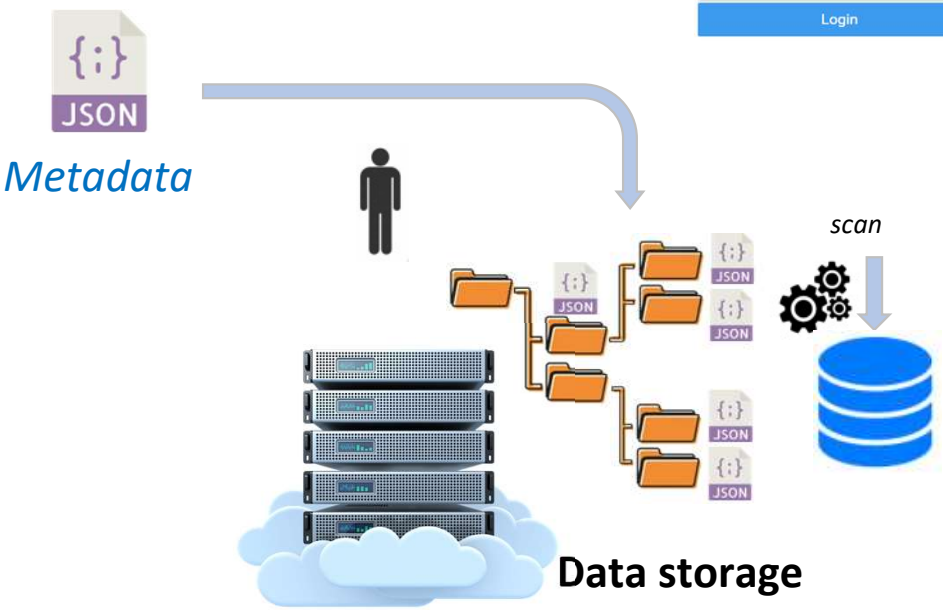


Maggot

Metadata aggregation on data storage

Possible **access** to data





Local (meta)data repository

Storage space becomes the data repository

Maggot

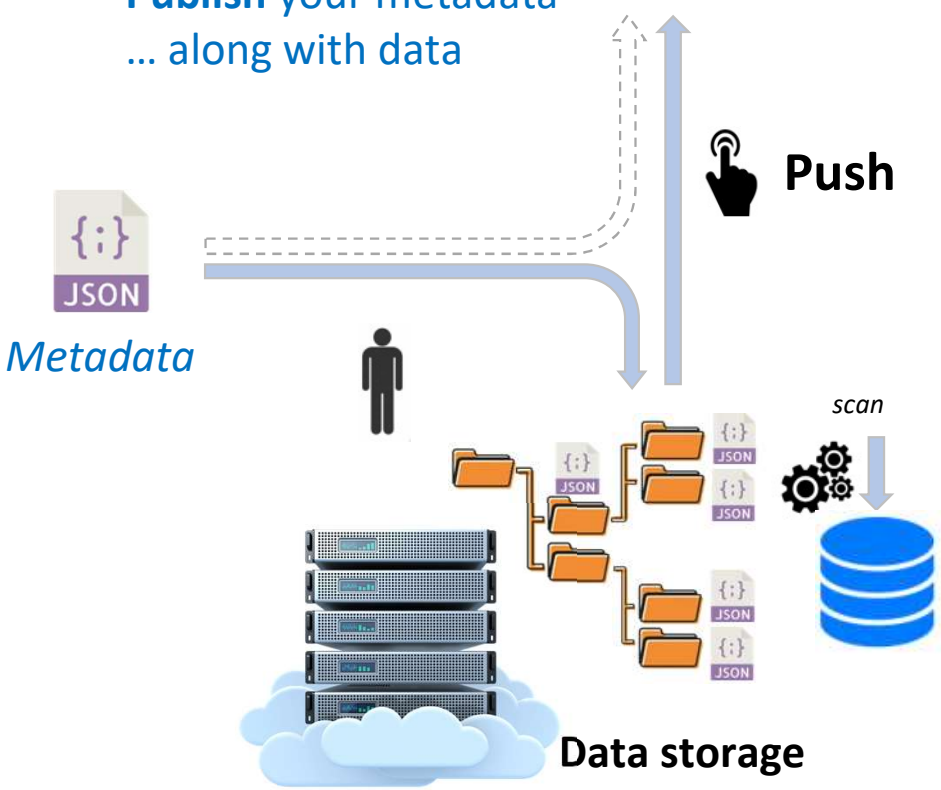
Metadata aggregation on data storage

Institutional data repositories

The Dataverse Project / zenodo

⇒ Have the privileges to do so (creation/modification rights).

Publish your metadata
… along with data



Local (meta)data repository

Storage space becomes the data repository

Citation Metadata

Dataset Persistent ID	doi:10.82233/FK2/3MHHX6	
Title	FRIM - Fruit Integrative Modelling	
Other ID	Maggot: frim1	
Contact	Use email button above to contact. Gibon Yves (INRAE)	
Author	Bénard Camille (INRAE) Biais Benoit (INRAE) Beauvoit Bertrand (Univ. Bordeaux) - ORCID: 0000-0002-7666-6429 Colombié Sophie (INRAE)	people CVLIST
Contributor	Data Collector : Bénard Camille (INRAE) Data Collector : Biais Benoit (INRAE) Data Collector : Ballias Patricia (INRAE) Data Collector : Maucourt Mickaël (Univ. Bordeaux) Data Curator : Moing Annick (INRAE) - ORCID: 0000-0003-1144-3600 Data Curator : Jacob Daniel (INRAE) - ORCID: 0000-0002-6687-7169 Project Leader : Gibon Yves (INRAE) - ORCID: 0000-0001-8161-1089 Work Package Leader : Vercambre Gilles (INRAE) - ORCID: 0000-0001-6486-9547	people CVLIST
Producer	Bordeaux Metabolome (INRAE) https://metabolome.cgfb.u-bordeaux.fr/	producer CVLIST
Language	English	producer CVLIST
Subject	Computer and Information Science; Medicine, Health and Life Sciences	producer CVLIST
Keyword	tomato http://purl.obolibrary.org/obo/NCBITaxon_4081 (EFO) fruit growth http://purl.obolibrary.org/obo/PO_0009001 (EFO) experimental measurement http://www.ebi.ac.uk/efo/EFO_0001444 (EFO)	bponto BioPortal
Topic Classification	fruit growth (thesaurus-inrae) http://opendata.inrae.fr/thesaurusINRAE/c_7655 plant health (thesaurus-inrae) http://opendata.inrae.fr/thesaurusINRAE/c_11833 omics (thesaurus-inrae) http://opendata.inrae.fr/thesaurusINRAE/c_e3728dc6 computer analysis (thesaurus-inrae) http://opendata.inrae.fr/thesaurusINRAE/c_16182	voinrae SKOSMOS
Data Type	Dataset	
Data Origin	experimental data	
Life cycle step	Study design; Data collection	
Related Publication	Biais B, Bénard C, Beauvoit B, Colombié S, Prodhomme D, Ménard G, Bernillon S, Gehl B, Gautier H, Ballias P, Mazat J-P, Sweetlove L, Génard M, Gibon Y. 2014. Remarkable reproducibility of enzyme activity profiles in tomato fruits grown under contrasting environments provides a roadmap for studies of fruit metabolism. Plant Physiology 164, 1204-1221 doi: 10.1104/pp.113.231241 https://doi.org/10.1104/pp.113.231241	
Other Reference	Experimental data tables: ODAM dataexplorer, https://pmb-bordeaux.fr/dataexplorer/?ds=frim1; Article: Beauvoit et al (2014) Plant Cell 26: 3224–3242 , https://doi.org/10.1105/tpc.114.127761; Article: Bénard et al (2015) Journal of Experimental Botany Vol. 66, No. 11 pp. 3391–3404, https://doi.org/10.1093/jxb/erv151; Article: Colombié el al (2015) Plant Physiology 180, 1709–1724 , https://doi.org/10.1104/pp.19.00086	
Funding Information	ANR: ANR-11-INBS-0010 ANR: ANR-11-INBS-0012	grant CVLIST
Depositor	Jacob Daniel	
Deposit Date	2023-04-04	

{ Mapping }

Maggot

Metadata aggregation on data storage

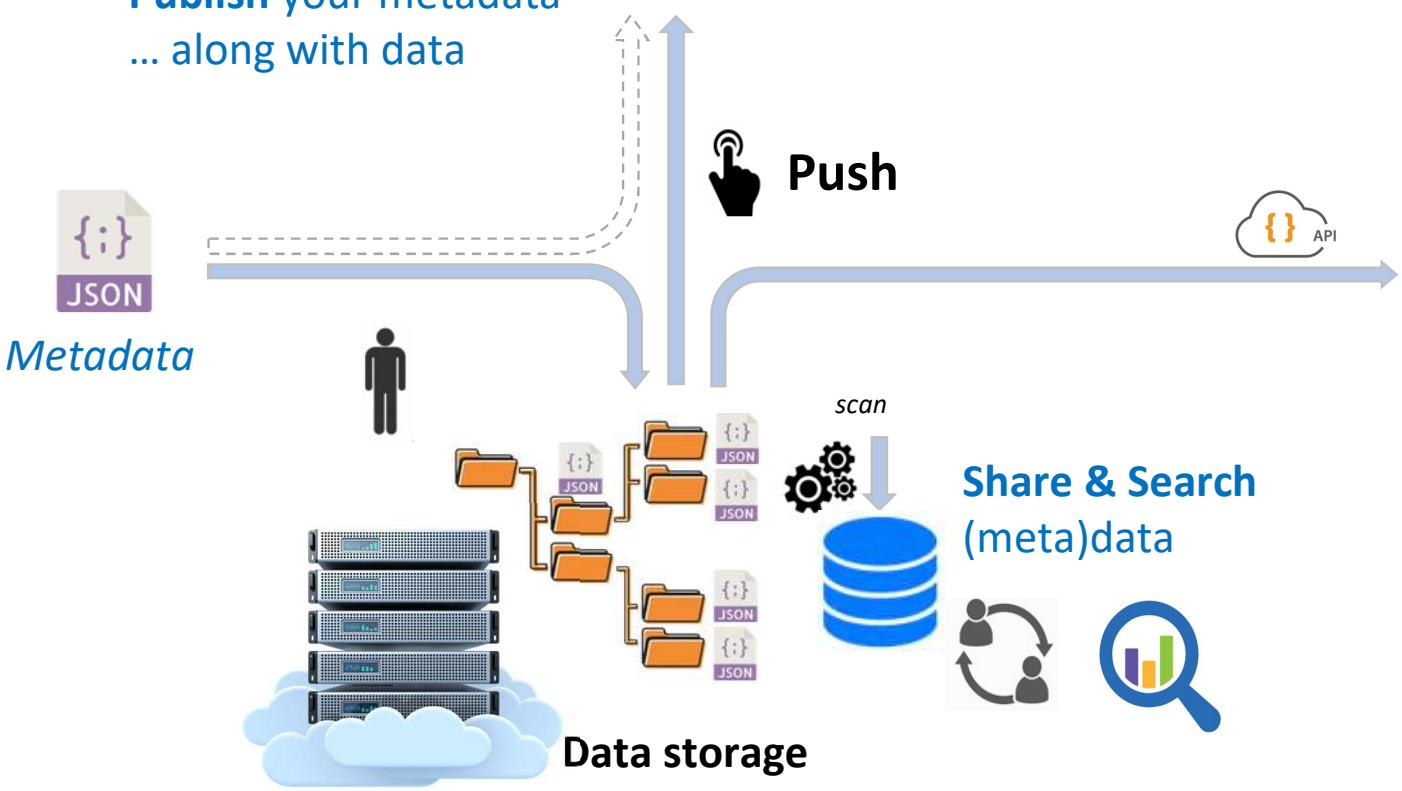
Institutional data repositories

The Dataverse Project / zenodo

Findable Accessible Interoperable Reusable **Interoperability**

Publish your metadata
… along with data

Allow machines to
collect metadata



Open Archives Initiative Protocol for Metadata Harvesting

XML *Dublin Core*

JSON-LD Linked Open Data

« Climb the LOD mountain»
gently, and step by step.

Local (meta)data repository

Storage space becomes the data repository

Maggot
Metadata aggregation on data storage

Configuration flexibility

Metadata management (local repository)

TSV Definition of terminology

TSV Metadata documentation

Metadata crosswalks (remote repository and/or metadata harvesting)

TSV Definitions of mapping
to Dataverse
To Zenodo

TSV Definitions of mapping
to JSON-LD
to OAI-PMH

What to describe & how to describe it

Metadata Schema DDI

Controlled Vocabulary

AgroPortal, BioPortal,
VOINRAE, LOTERRE, ONTOSTACK, …

Repositories & export formats

The Dataverse Project

zenodo

JSON-LD

…

Maggot
Metadata aggregation on data storage

Roles

All data stakeholders must be involved according to their « job skills ».

Standards & Terminologies
The 4 **FAIR** pillars

DDI
Data Document Initiative

OntoPortal
(AgroPortal, BioPortal, …)

Thesaurus SKOSMOS
(VOINRAE, LOTERRE, ONTOSTACK, …)

Roles

Upstream

Fix the frame based on standards and vocabularies recognized by the international community in the field.

Guidelines, Terminologies

Data manager
+
Data producers

Metadata Capture

« Input Mask »

Allowing metadata to be captured according to a data description standard

Good Practices

Identifiers, Terminologies

Data producers

Downstream

Transform data for interoperability

greatly facilitated by the task upstream

The Dataverse Project, zenodo, JSON-LD, …

Guidelines, Terminologies, Identifiers, Formats

Data manager

Maggot

Metadata aggregation on data storage

The Maggot tool allows a collective to :

- **Have visibility** of what is produced within the collective
  - datasets, software, databases, images, sounds, videos, analyses, codes, ...
  - ⇒ **share metadata**

- **Raise awareness** among newcomers and students about a better description of what they produce
  - Limit data loss (after temporary staff leave)

- **Promote FAIR** within the collective
  - particularly as part of a quality approach

https://pmb-bordeaux.fr/maggot/

https://inrae.github.io/pgd-mmdt/

Meet Open Data requirements

This is not necessarily making data to open access without conditions, but rather

1. **provide access to metadata** defining the conditions of access and use of data,
2. **open the data beyond itself,** i.e. that they can be interoperable.

So the data must be **as FAIR as possible**, …
…even if it is not possible to make them open.